Title

Evaluating South Africa's Regulatory Readiness for Integrating Software as a Medical Device (SaMD) and AI as a Medical Device (AIaMD): A Comprehensive Gap Analysis and Stakeholder-Driven Recommendations

Background and Motivation

The emergence of advanced digital health technologies, notably Software as a Medical Device (SaMD) and AI-driven medical devices (AIaMD), offers significant potential to revolutionize healthcare delivery in South Africa. These technologies enhance diagnostic precision (e.g., AI detection of large vessel occlusions in stroke care, Elijovich et al., 2020), improve accessibility (e.g., deep-learning-guided echocardiograms, Narang et al., 2021), and optimize clinical workflows (e.g., AI in breast cancer screening, Shoshan et al., 2020). However, their integration is challenged by a regulatory framework under the South African Health Products Regulatory Authority (SAHPRA) that is predominantly tailored to traditional, hardware-based medical devices. Key issues such as continuous software updates, cybersecurity risks, algorithmic bias, and data privacy remain insufficiently addressed, posing barriers to safe and ethical deployment.

Problem Statement

South Africa’s current regulatory landscape, anchored in the Medicines and Related Substances Act 101 of 1965 and SAHPRA’s Medical Device Guidelines (2021, 2023), lacks the granularity to govern SaMD and AIaMD effectively. Critical dimensions—including algorithmic transparency, accountability, bias mitigation, technical robustness, and post-market surveillance—are either absent or inadequately specified. This regulatory gap threatens patient safety, data integrity, and trust in these technologies, limiting their adoption and innovation potential within the South African healthcare ecosystem.

Research Aim

To rigorously evaluate South Africa’s regulatory preparedness for SaMD and AIaMD adoption through a multi-faceted approach involving a gap analysis of existing regulations, stakeholder consultations, and the formulation of technically informed, actionable recommendations to address identified deficiencies.

Research Objectives

1. **Literature Review**: Conduct an in-depth review of global regulatory frameworks and technical standards for SaMD and AIaMD to establish a baseline of best practices.

2. **Governance Criteria Development**: Define a comprehensive, technically grounded set of criteria for regulators to evaluate SaMD and AIaMD submissions, emphasizing algorithmic integrity, cybersecurity, and ethical deployment.
3. **Gap Analysis**: Assess South Africa's regulatory framework against the developed criteria to pinpoint specific shortcomings.
4. **Stakeholder Questionnaire**: Design and deploy a questionnaire to capture awareness and perceptions of regulatory gaps among key stakeholders—IRBs, SAHPRA, SAMRC, CROs, and universities.
5. **Analysis and Recommendations**: Synthesize findings from the gap analysis and stakeholder input to propose precise, implementable recommendations for regulatory enhancement.

Research Questions

- What are the essential technical and ethical criteria for governing SaMD and AIaMD that regulators must prioritize?
- What specific deficiencies exist in South Africa's regulatory framework for SaMD and AIaMD compared to international benchmarks?
- How do stakeholders (IRBs, SAHPRA, SAMRC, CROs, universities) perceive these gaps, and what solutions do they propose?
- What technically feasible strategies can address these gaps to ensure safe, ethical, and effective deployment of SaMD and AIaMD in South Africa?

Methodology

1. Literature Review

- **Scope**: Analyze global regulatory frameworks and technical standards, including:
    - U.S. FDA's *Good Machine Learning Practice for Medical Device Development* (2021) and Predetermined Change Control Plan (PCCP).
    - EU *Regulation 2017/745 (MDR)* for lifecycle management and clinical evaluation.
    - ISO 13485:2016 (quality management systems) and IEC 62304:2006 (software lifecycle processes).
    - WHO *Global Model Regulatory Framework for Medical Devices* (2017).
- **Focus Areas**: Extract principles related to accountability, transparency, fairness, human oversight, technical robustness, cybersecurity, data privacy, and societal impact.
- **Sources**: Peer-reviewed studies (e.g., Chen et al., 2020 on explainable AI; Challen et al., 2019 on bias mitigation) and industry practices (e.g., Scarlet.cc certification).

2. Governance Criteria Development

- **Approach**: Synthesize findings into a detailed list of evaluation criteria, including:
    - **Algorithmic Transparency**: Requirements for explainability (e.g., XAI techniques per Chen et al., 2020) and documentation of decision-making processes.

    - **Data Privacy**: Compliance with POPIA (South Africa) and GDPR-like standards for data handling and consent.
    - **Bias Mitigation**: Strategies for detecting and correcting algorithmic bias (e.g., Challen et al., 2019).
    - **Cybersecurity**: Protocols for securing software against threats (e.g., ISO/IEC 27001 integration).
    - **Continuous Monitoring**: Post-market surveillance and drift detection mechanisms (e.g., Bayram & Alatas, 2021).
    - **Validation/Verification**: Clinical and technical validation per IEC 62304:2006.
    - **Ethical Oversight**: Mechanisms for human-in-the-loop governance (Papagiannidis et al., 2025).
- **Output**: A checklist for regulators to assess SaMD/AIaMD submissions.

3. Gap Analysis

- **Baseline**: South Africa's Medicines and Related Substances Act 101 of 1965 and SAHPRA Guidelines (2021, 2023).
- **Method**: Map these regulations against the governance criteria to identify absences or weaknesses, e.g., lack of provisions for adaptive algorithms (FDA PCCP) or cybersecurity mandates.
- **Tools**: Comparative matrices and qualitative assessments.

4. Stakeholder Questionnaire

- **Design**: Develop a structured questionnaire with Likert-scale and open-ended questions aligned with governance criteria (e.g., "How adequately does SAHPRA address algorithmic bias?").
- **Participants**: IRBs, SAHPRA officials, SAMRC researchers, CROs, and university academics.
- **Distribution**: Online surveys and semi-structured interviews for qualitative depth.
- **Analysis**: Quantitative (statistical trends) and qualitative (thematic coding) methods.

5. Analysis and Recommendations

- **Synthesis**: Cross-reference gap analysis with stakeholder feedback to validate findings and identify priorities.
- **Recommendations**: Propose specific regulatory updates, e.g., adopting PCCP-like frameworks for software updates, mandating bias audits, or establishing a SaMD/AIaMD-specific SAHPRA task force.
- **Technical Detail**: Include implementation steps, such as integrating IEC 62304 processes or Scarlet.cc-like certification pathways.

Expected Outcomes

- A technically robust list of governance criteria for SaMD and AIaMD evaluation.

- A detailed gap analysis report outlining deficiencies in South Africa’s regulatory framework.
- Stakeholder-informed insights into regulatory awareness and needs.
- A set of precise, actionable recommendations to enhance regulatory clarity and safety.

Significance of the Study

This research addresses a pressing need to modernize South Africa’s regulatory framework for emerging health technologies. By closing gaps, it will enhance patient safety, foster ethical AI deployment, and support innovation among startups, positioning South Africa as a leader in the global digital health arena.

Scope

The study focuses on the regulatory governance of SaMD and AIaMD within South Africa, targeting the perspectives of IRBs, SAHPRA, SAMRC, CROs, and universities. It excludes technical development or clinical efficacy assessments, concentrating solely on regulatory pathways and oversight mechanisms.